SIMON FRASER UNIVERSITY
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WorkSafeBC: Modelling Industry-Specific Claim Costs

Prepared by Stuart Siu, Peter Bui, and Anna Lukanov

Introduction & Objective

This project analyzes how WorkSafeBC's total claim costs evolve and the extent to which both macroeconomic conditions and industry-specific risk factors influence them.

The primary objective is to model payroll-adjusted claim costs, *or pure premium,* while:

- isolating the underlying injury-year time trend, and
- accounting for heterogeneity across industry subsectors.

To achieve this, a hierarchical modelling framework is employed to separate overall time effects from subsector-level variation. This approach allows us to assess whether claim cost growth is consistent across industries or varies systematically within subsectors of British Columbia's labour market.

Data Source

Data Extraction

The Industry Health & Safety dataset from WorkSafeBC is accessed through an interactive Power BI dashboard, which does not allow direct data export at the required level of detail. To overcome this limitation, an automated data extraction process was developed using Selenium.

This process systematically retrieves the required data from the dashboard, after which it is parsed, cleaned, and transformed into a structured long format suitable for statistical modelling and analysis.

Data Description

The dataset consists of annual observations at the Classification Unit (CU) level across British Columbia, covering the period from 2015 to 2024. It includes total claim costs and assessable payroll, along with a breakdown of claim costs into:

- current-year (injury year) claims, and
- prior-year (other years) claims.

This distinction captures both newly incurred claims and the ongoing development of existing claims, providing a more comprehensive view of how claim costs emerge and evolve.

For the first phase of the analysis, the data are aggregated to a broader subsector level to establish stable, high-level trends while maintaining sufficient variation for modelling.

Phase 1: Subsector Level

Motivation for Modelling Approach

Observed claim costs are not directly comparable across industries due to differences in payroll exposure. Even after normalization, sector-level aggregation can obscure substantial variation across subsectors. As shown in **Figure A-1**, the shaded bands indicate considerable within-sector variability, highlighting the extent of heterogeneity that is not captured by sector averages alone.

To prevent this heterogeneity from being masked, Phase 1 applies our hierarchical modelling framework to isolate subsector-specific variation from general time trend.

Key Variables

The dataset includes several variables relevant to modelling claim costs:

- Response Variable: Total claim costs
- Exposure Variable: Payroll
- Primary Predictor: Injury-year (to capture the time trend)
- Grouping Variable: Industry subsector

Payroll is incorporated as an offset in the modelling framework to estimate pure premium, enabling meaningful comparisons across industries with different exposure levels.

Model Selection

A generalized linear model (GLM) is used as a baseline, assuming independence across observations and a common structure across all subsectors. However, this assumption is unlikely to hold in the context of workers' compensation data, where industries exhibit inherent structural differences in risk. The model is therefore extended to a generalized linear mixed model (GLMM), allowing subsector-specific variation in both baseline risk levels and time trends through random effects.

$$\log\left(Cost_{i,t}\right) = \log\left(Payroll_{i,t}\right) + (\beta_0 + u_i) + (\beta_1 + v_i) \cdot Year_t$$

$\beta_0 + \beta_1 \cdot Year_t$ is the fixed effect;

$u_i + v_i \cdot Year_t$ is the subsector-specific random effect.

All models are estimated using maximum likelihood methods. The GLMM is implemented in R using the *lme4* package, with random effects integrated via the Laplace approximation.

Data Splitting

To evaluate model performance, a time-based train-test split is applied. Observations from 2015 to 2022 are used to fit the model, while data from 2023 to 2024 are held out for testing.

Claim Type Separation

A key limitation of using aggregated claim cost data is that current-year observations do not fully capture the complete development of claims over time. Current-year costs primarily reflect newly reported incidents and initial cost estimates, while excluding subsequent adjustments, developments, and ongoing payments. To address this, total claim costs are split into current-year and prior-year components, as illustrated in **Figure A-2**. This decomposition allows the model to better align claim costs with their underlying development patterns. By explicitly modelling prior-year development, the model captures both new claims and the evolution of existing claims, resulting in a more accurate representation of total claim costs.

Trend Stability

Trend analysis indicates a break in the data structure. From 2015 to 2019, claim cost trends were relatively stable across subsectors. However, starting in 2020, volatility and divergence across sectors increased, likely due to external shocks such as COVID-19. Since the dataset has only 10 annual observations, it is challenging to determine if this signifies a lasting structural change or a temporary disruption. Consequently, fluctuations after 2020 are viewed cautiously as short-term variations rather than signs of a sustained trend change. Higher-frequency data, such as quarterly observations, would better distinguish short-term shocks from long-term trends.

Exposure Alignment

For current-year claims, current payroll is an appropriate exposure measure. However, for prior-year runoff claims, a mismatch arises between exposure (payroll) and claim development timing. Payroll reflects current economic activity, whereas claim costs may emerge over multiple periods. As shown in **Figure A-4**, different weighted averages of lagged exposure produce similar linearity with prior-year claim costs. To optimize this, we implemented a bifurcated GLMM framework that separately models current-year and prior-year claims with their best match payrolls. This improves alignment between exposure and claim development, enhancing both model accuracy and interpretability.

Model Evaluation Criteria

Model performance is evaluated using Root Mean Squared Predictive Error (RMSPE) on the test dataset. This ensures that model selection is based on out-of-sample predictive performance rather than in-sample fit. Akaike Information Criterion (AIC) was not used due to differences in model structure, which make it less appropriate for comparison.

Results Overview

Model performance improves as complexity increases. The baseline GLM exhibits the highest prediction error, while the GLMM significantly reduces error by capturing unobserved heterogeneity across subsectors. Further improvements are achieved through claim-type separation and exposure alignment. In particular, the bifurcated GLMM, which incorporates 1-year lagged payroll on prior-year claims, delivers the strongest performance with the lowest predictive error, as shown in **Figure A-5**. These results highlight the importance of both aligning exposure with claim development and accounting for cross-sectional differences and temporal dynamics. Additional model comparison results are summarized in **Figure B-1** in Appendix B.

Empirical analysis further indicates that a one-year lag specification exhibits the strongest relationship between log claim costs and log payroll, providing the best fit among the alternatives. This suggests that claim development is more closely associated with prior-year exposure rather than other years' payroll. Incorporating lagged exposure (payroll), therefore, improves both interpretability and predictive accuracy by better aligning the risk exposure with the timing of claim realization.

Motivation for Classification Unit-Level Modelling

Despite these improvements, the model retains an error margin of approximately 11.83%, suggesting that significant variability remains within specific industries**. Figure B-2** in Appendix B highlights the variation in risk and cost patterns across sectors, with certain industries clearly deviating from the overall trend. **Figure B-3** further summarizes subsector-level risk trajectory based on changes in new claim frequency and prior-year cost development. While the subsector-level model captures these broad differences, it still aggregates over potentially diverse underlying activities within each subsector.

This motivates a more granular analysis at the Classification Unit level, where within-subsector variation can be more accurately captured, and model performance further improved.

Phase 2: Classification Units (CU) Analysis

Building on the macroeconomic-level time trends and optimal exposure alignment established in Phase 1, Phase 2 shifts focus on the structural disparities within individual industries. This phase aims to identify specific occupational hazards by modelling at the more granular CU level. To accomplish this, relying solely on payroll and time data is inadequate, so we must incorporate other operational risk drivers from the available data. This involves converting raw claim counts into standardized, direct risk metrics, specifically using time-loss claims as a proxy for claim frequency, and average work days lost as an indicator of claim severity.

Feature Selection

An iterative backward stepwise GLM, implemented using the *stats* package in R, is used to select the relevant predictors. This approach systematically removes non-significant variables, retaining only the most statistically significant cost drivers for both Injury Year and Prior Year models.

Structural Isolation

The selected features are then incorporated into a final GLMM with CU included as a random intercept. The resulting variance captured by this random effect can be interpreted as the structural penalty or discount.

Analysis Results

The following results highlight the top three most volatile subsectors, illustrating how cost drivers and structural risk vary across industries at a more granular level.

Subsector 7020 – Fishing

Injury Year

At the CU level within the Fishing subsector, injury-year costs are primarily driven by claim frequency, as measured by time-loss claims. An increase of 10 time-loss claims is associated with approximately an 18% increase in injury-year costs, holding structural differences constant. This reflects frequency-driven risk linked to day-to-day fishing operations. Higher early return-to-work (RTW) rates are also associated with lower injury-year costs, highlighting the importance of early intervention in mitigating initial claim severity, as shown in **Figure C-7**.

Prior Year

In contrast, long-term costs are driven more by claim severity and duration than by frequency. An increase of 100 days in average claim duration is associated with roughly a 17% increase in runoff costs, illustrating the long-tail nature of fishing-related injuries.

Variability in RTW outcomes remains a significant factor, as delayed RTW prolongs claim duration and amplifies cost development. Notably, the impact of claim frequency diminishes, and even reverses, in the prior-year model, indicating that persistent claims costs are primarily driven more by severity than by frequency.

CU Variance

Risk varies substantially across fishing methods, with CU-level random effects confirming that these differences are structural. More hazardous operations, such as seine and trawl fishing, show positive deviations in both injury-year and prior-year components, indicating higher baseline risk and long-tail exposure. This is supported by large positive coefficients for these categories in the fixed-effects model and further illustrated in **Figure C-4**.

In contrast, fin fish farming shows consistently negative deviations, reflecting a structurally lower-risk profile. More stable operating conditions support earlier RTW, shorter claim durations, and lower overall costs.

Subsector 7220 – Heavy Construction

Injury Year

Immediate costs are heavily driven by raw injury frequency (**Figure C-2**), with time loss claims acting as a highly significant multiplier (approximately 1.04x). While the initial model suggested an early return-to-work ratio, subsequent ANOVA testing proved this to be a structural illusion. This confirms that the frequency of minor injuries is not an independent driver, but rather a specific nature of some construction trade.

Prior Years

The ANOVA test results also indicated that claim frequency does not have significant predictive power on prior-year costs, meaning that runoff claims are entirely hard-coded into the structural severity within the specific industry. This reinforces the distinction observed in other subsectors, where long-term costs are dominated by severity rather than frequency.

CU Variance

By isolating this structural variance (**Figure C-5**), the model clearly divides the subsector into manual and mechanized trades. Manual trades, such as iron workers and welders, are exposed to extreme heights and heavy lifting, resulting in both higher upfront and more severe long-term recovery costs. Conversely, mechanized operations, such as heavy equipment operators from tunnel construction and pile driving, are safer, with injuries most commonly due to overexertion and repetitive motion. (WorkSafeBC, 2025)

Some CUs experienced explosive payroll growth in post-2020, due to large-scale provincial infrastructure projects. This extreme wage inflation can distort pure premium estimates and therefore drag down the risk trend. An average wage variable (*avg_wage*) was also included as a candidate feature; however, it was not retained as the stepwise algorithm rejected it. This mathematically proves that our 10-year baseline and CU variables successfully mitigated the structural anomaly from the 3-year length mega-projects.

Subsector 7530 – Public Administration

Injury Year

Consistent with other subsectors, overall claim frequency remains the universal driver of immediate injury-year costs as shown in **Figure C-3**. An increase of 100 claims is associated with approximately a 4% increase in costs, indicating a relatively moderate sensitivity to frequency compared to other industries.

Prior Years

In the runoff period, the predictive power of frequency diminishes, and severity becomes the dominant driver. An increase of 10 average days lost per claim is associated with an approximate 7% increase in long-term costs. This relationship holds across different roles within the subsector, suggesting that regardless of the work environment, whether administrative or field-bases, such as on police patrol, the severity of the injuries is still the clear driver of long-term cost development.

CU Variance

The structural risk chart (**Figure C-6**) reveals clear differences across the three Classification Units. First Nations Operations exhibit the lowest risk levels, with consistently lower pure premiums after isolating the frequency and severity impact. Local government roles show moderate risk, with relatively higher impact on long-term claims.

In contrast, Law Enforcement demonstrates significantly higher risk, with substantial deviations from the other two CUs in both injury-year and prior-year costs. When the "Acts of Violence" (*aov_rate*) variable is introduced, it initially appears highly significant through the stepwise algorithm. However, once CU-level variables are included, this significance is completely absorbed by the Law Enforcement category. This indicates that exposure to violent incidents is inherently embedded within the structural risk profile of Law Enforcement, rather than acting as an independent explanatory variable.

Recommendations

The primary objective of isolating time trends and structural variance is to enable public sectors and industry safety associations to identify the true underlying drivers of claim costs. Based on our empirical findings, we recommend the following approaches to risk evaluation and prevention:

- **Bifurcated risk mitigation:** The statistical failure of claim frequency to predict prior-year runoff costs indicates that day-one risks and long-tail risks require different prevention strategies. Risk management efforts should be tailored to the component driving cost deterioration.
- **Identify true occupational hazards:** Without controlling macroeconomic inflation and exposure levels, the true underlying risk drivers of an industry may remain hidden. Structural isolation allows for prevention resources to be allocated to where they are most needed.
- **Adopt extended evaluation horizons:** Short-term assessments are vulnerable to temporary shocks, while a decade-long baseline smooths these effects and reveals the underlying structural trend of each subsector.

Limitations and Future Improvements

1) **Accident year and development data:** The absence of accident-year information and detailed loss development data limits the ability to precisely align claims with their originating exposure periods. As a result, prior-year payroll must be approximated using a rolling assumption, which may introduce some distortion into trend estimates.
2) **Data frequency:** The use of annual data restricts the ability to capture short-term dynamics and seasonal patterns in claims. This reduces the model's sensitivity to within-year variations that may be important in industries with cyclical or weather-dependent risk profiles.
3) **Level of aggregation:** The analysis is based on aggregate frequency and severity measures without further segmentation by injury type, occupation, or risk classification. Incorporating more granular injury-level data would likely improve explanatory power and provide more targeted insights into the underlying drivers of claim costs.

Despite data limitations, the strength of this analysis lies in its top-to-bottom architecture. By establishing macroeconomic baselines first and subsequently fine-tuning features for specific subsectors, this framework remains transparent. It provides policymakers and the industries with a targeted, interpretable tool that highlights exactly where and how to deploy prevention resources most effectively.

Appendix A – Data Pre-Analysis

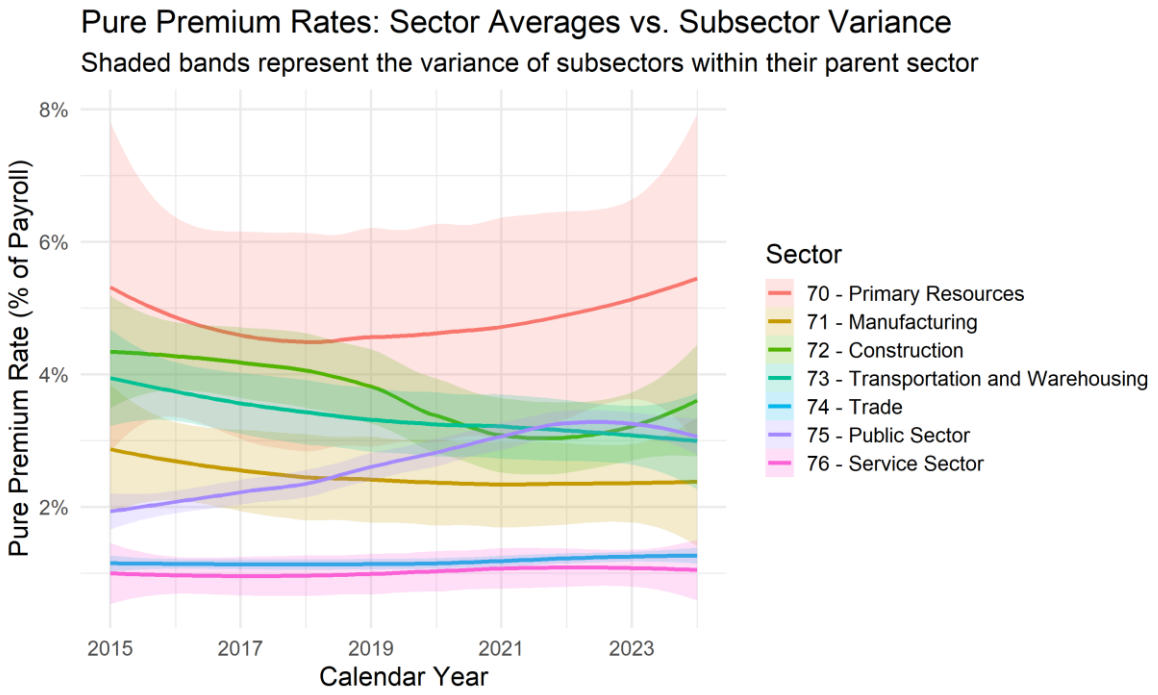

Figure A-1: Sector-Level Pure Premium Rates with Subsector Variability Over Time

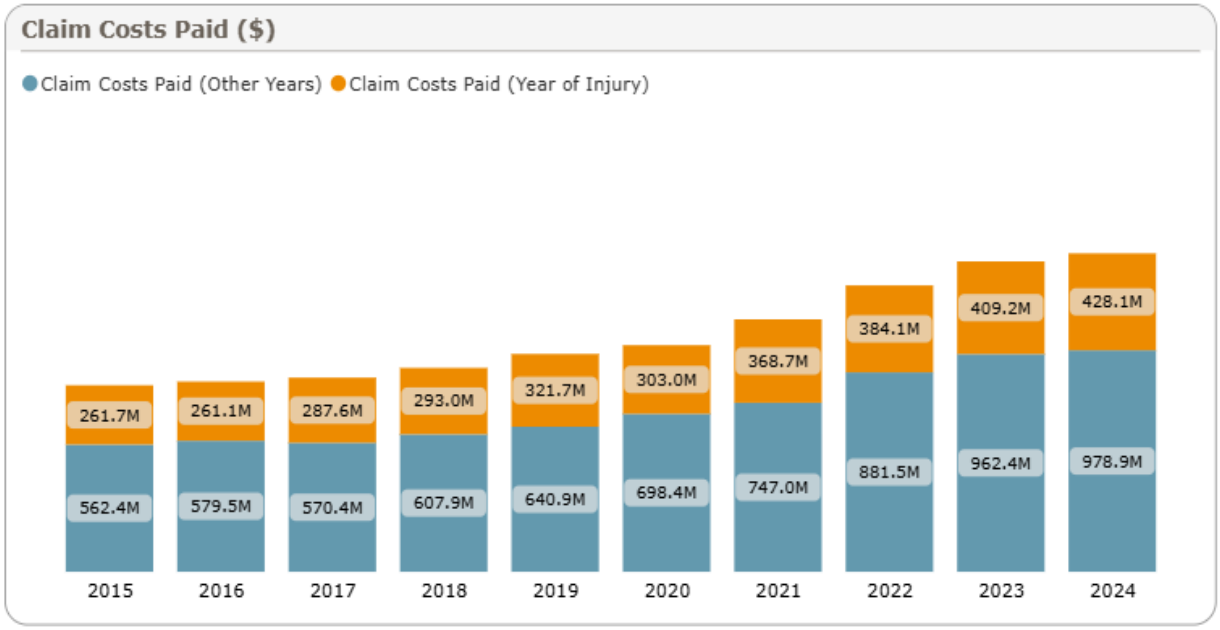

Figure A-2: Annual Claim Costs by Injury Year and Prior Year Components
Source: (WorkSafeBC, 2025)

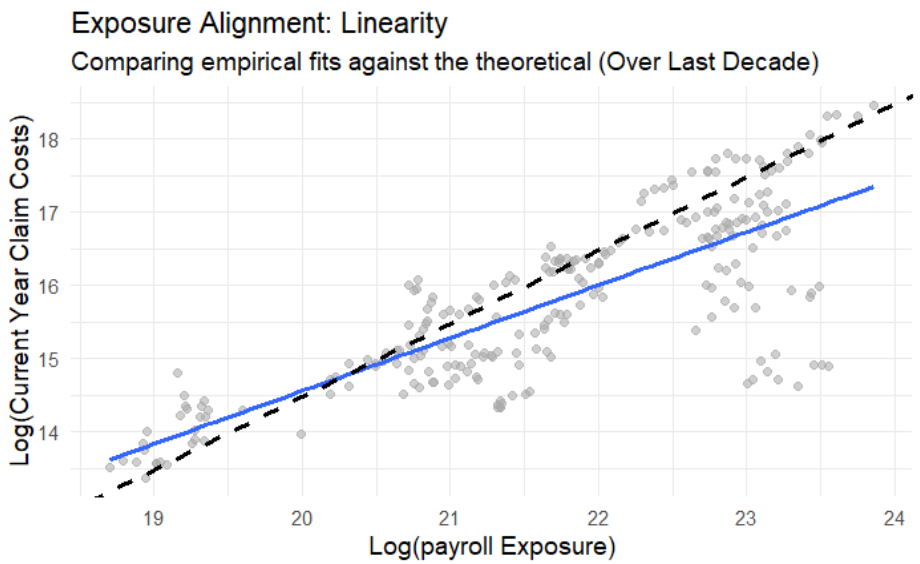

Figure A-3: Prior-Year Prediction Error under Alternative Exposure Weighting Schemes

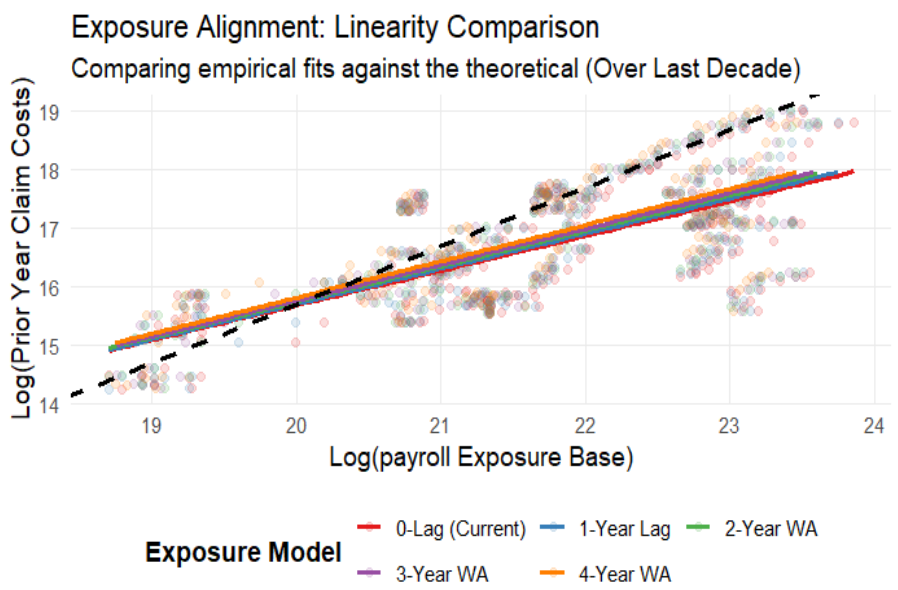

Figure A-4: Linearity of Log Claim Costs with Respect to Log Payroll Exposure

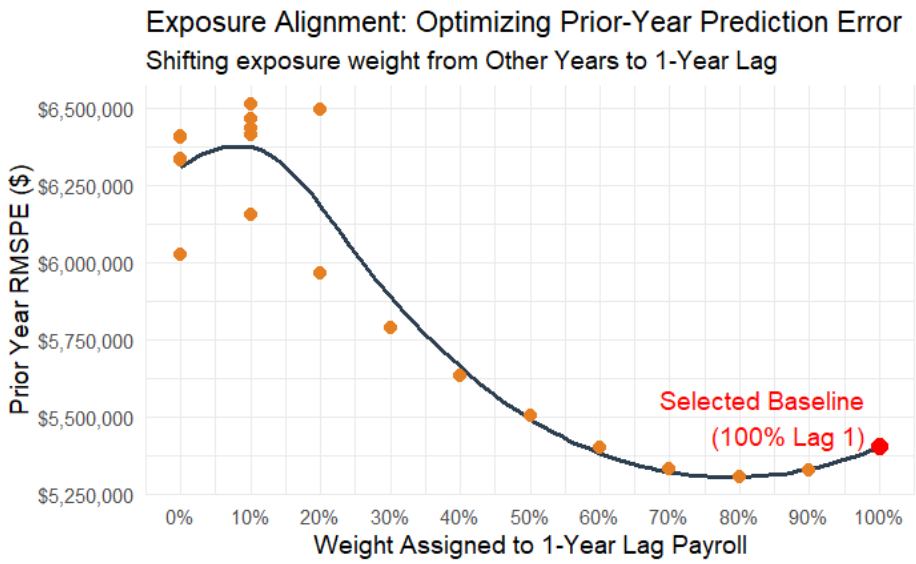

Figure A-5: Relationship Between Log Prior-Year Claim Costs and Log Payroll Exposure Across Lag Specifications

Appendix B – Empirical Results

Final Model Evaluation Summary

Architecture	Model Type	Exposure Base	8-Year Fit (RMSE)	Test Set (RMSPE)	Claim Cost Error Margin
Single	Fixed	Lag 0	$8,022,232	$12,481,311	21.56%
Single	Mixed	Lag 0	$4,419,811	$8,861,864	15.31%
Bifurcated	Fixed	IY: Lag 0 / OY: Lag 1	$5,162,637	$13,887,321	23.99%
Bifurcated	Mixed	IY: Lag 0 / OY: Lag 0	$4,527,384	$8,065,182	13.93%
Bifurcated	**Mixed**	**IY: Lag 0 / OY: Lag 1**	**$4,284,464**	**$6,847,134**	**11.83%**
Bifurcated	Mixed	IY: Lag 0 / OY: Lag 2	$4,430,322	$8,013,351	13.84%

Figure B-1: Five-Year Risk Trajectory by Sector

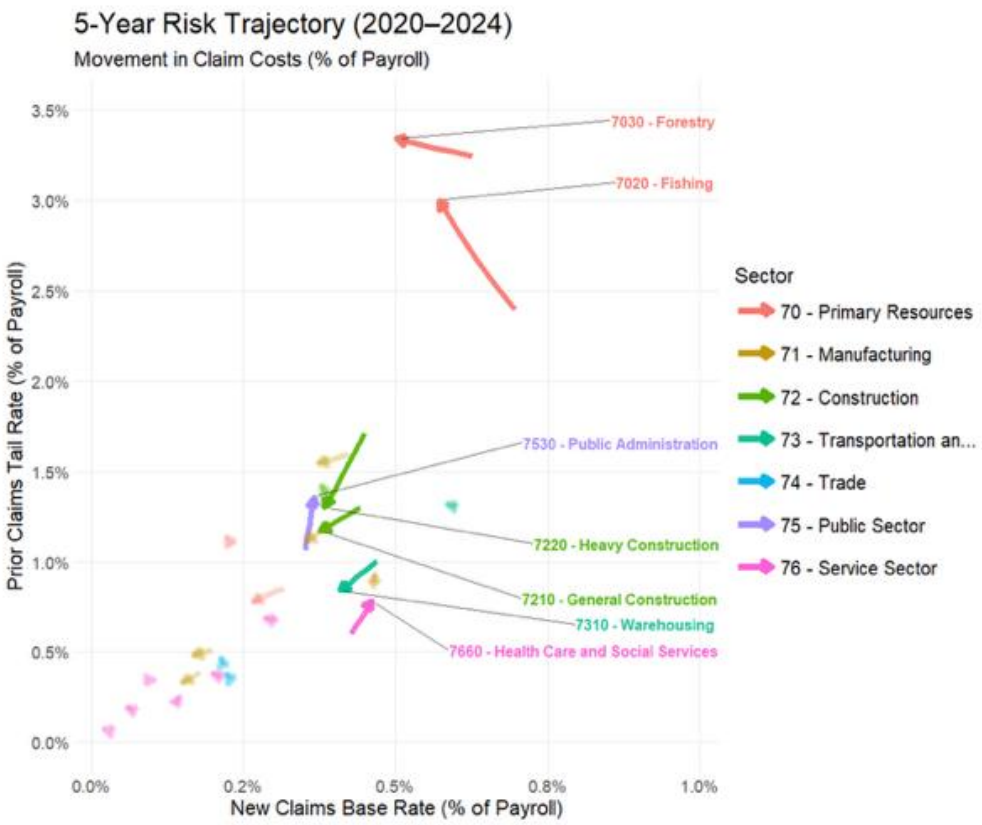

Figure B-2: Final Model Evaluation Summary of Predictive Performance Across Models

Top Risk Trajectories by Subsector

Subsector	trend_new	trend_prior	Vector_Length	Rate_Drift
7020 - Fishing	-4.8%	5.8%	0.62%	0.48%
7220 - Heavy Construction	-3.9%	-6.7%	0.42%	0.48%
7530 - Public Administration	1.1%	6.3%	0.29%	0.31%
7660 - Health Care and Social Services	1.9%	6.9%	0.19%	0.22%
7310 - Warehousing	-3.4%	-4.3%	0.17%	0.22%
7030 - Forestry	-5.4%	0.7%	0.16%	0.03%
7210 - General Construction	-3.9%	-2.6%	0.14%	0.19%

Figure B-3: Subsector Risk Trajectories Based on Changes in New Claim Frequency and Prior-Year Cost Development

Appendix C – CU Analysis

Fishing: GLMM Cost Drivers						
	Injury Year (IY)			Prior Year (OY)		
Characteristic	**exp(Beta)**	**95% CI**	**p-value**	**exp(Beta)**	**95% CI**	**p-value**
(Intercept)	0.886	0.511, 1.54	0.7	1.12	0.576, 2.18	0.7
I(time_loss_claims/10)	1.18	1.00, 1.40	**0.046**			
I(avg_days_lost/100)				1.17	1.01, 1.36	**0.039**
Abbreviation: CI = Confidence Interval						

Figure C-1: GLMM Cost Driver Estimates for Injury Year and Prior Year (Fishing Subsector)

Heavy Construction: GLMM Cost Drivers								
	Injury Year (IY)				Prior Year (OY)			
Characteristic	**Group**	**exp(Beta)**	**95% CI**	**p-value**	**Group**	**exp(Beta)**	**95% CI**	**p-value**
(Intercept)	cond	0.466	0.311, 0.696	<**0.001**	cond	0.980	0.719, 1.34	0.9
time_loss_claims	cond	1.04	1.02, 1.06	<**0.001**				
Abbreviation: CI = Confidence Interval								

Figure C-2: GLMM Cost Driver Estimates for Injury Year and Prior Year Claim Costs (Heavy Construction Subsector)

Public Admin: GLMM Cost Drivers						
	Injury Year (IY)			Prior Year (OY)		
Characteristic	**exp(Beta)**	**95% CI**	**p-value**	**exp(Beta)**	**95% CI**	**p-value**
(Intercept)	0.689	0.304, 1.56	0.4	0.682	0.247, 1.89	0.5
I(time_loss_claims/100)	1.04	1.00, 1.08	**0.047**			
I(avg_days_lost/10)				1.07	1.00, 1.14	**0.047**
Abbreviation: CI = Confidence Interval						

Figure C-3: GLMM Cost Driver Estimates for Injury Year and Prior Year Claim Costs (Public Administration Subsector)

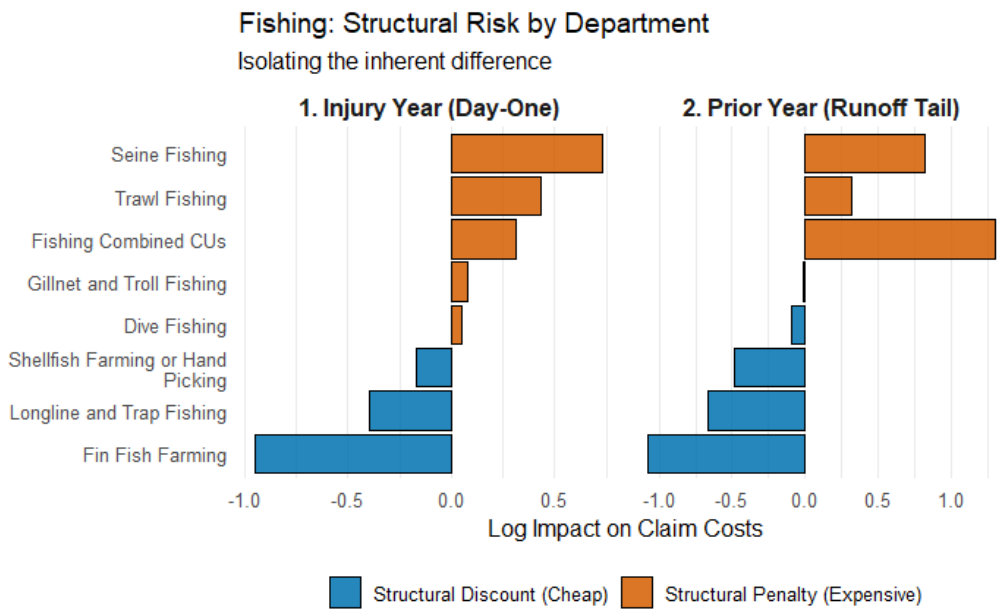

Figure C-4: Structural Risk Differences Across Fishing Departments for Injury Year and Prior Year Costs

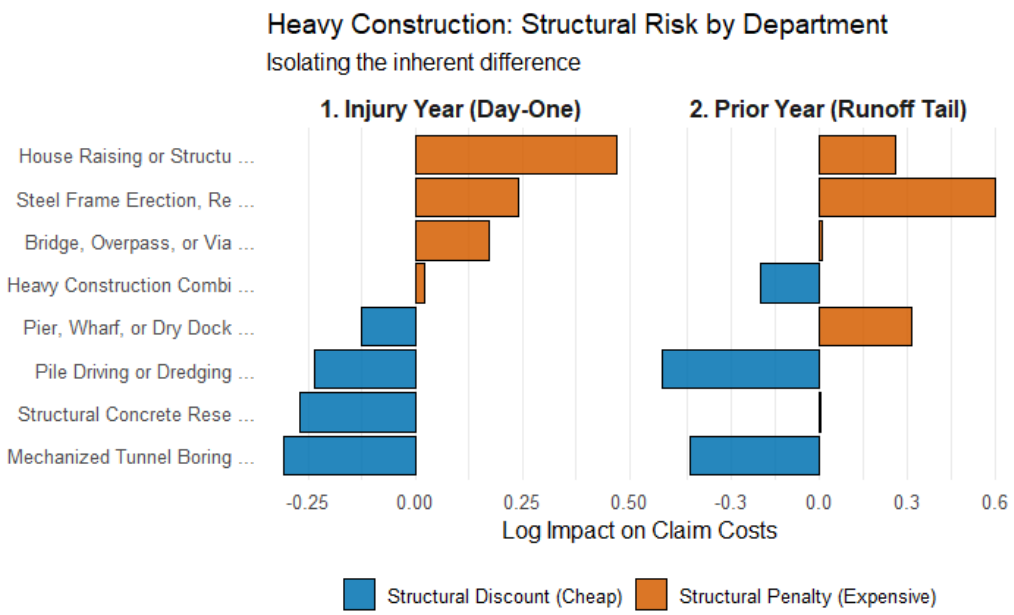

Figure C-5: Structural Risk Differences Across Heavy Construction Departments for Injury Year and Prior Year Costs

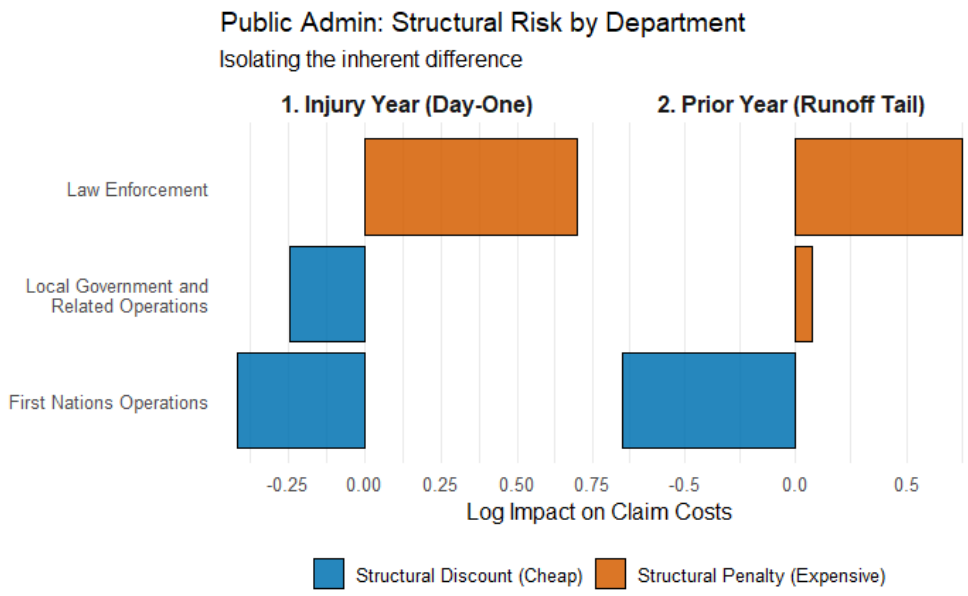

Figure C-6: Structural Risk Differences Across Public Administration Departments for Injury Year and Prior Year Costs

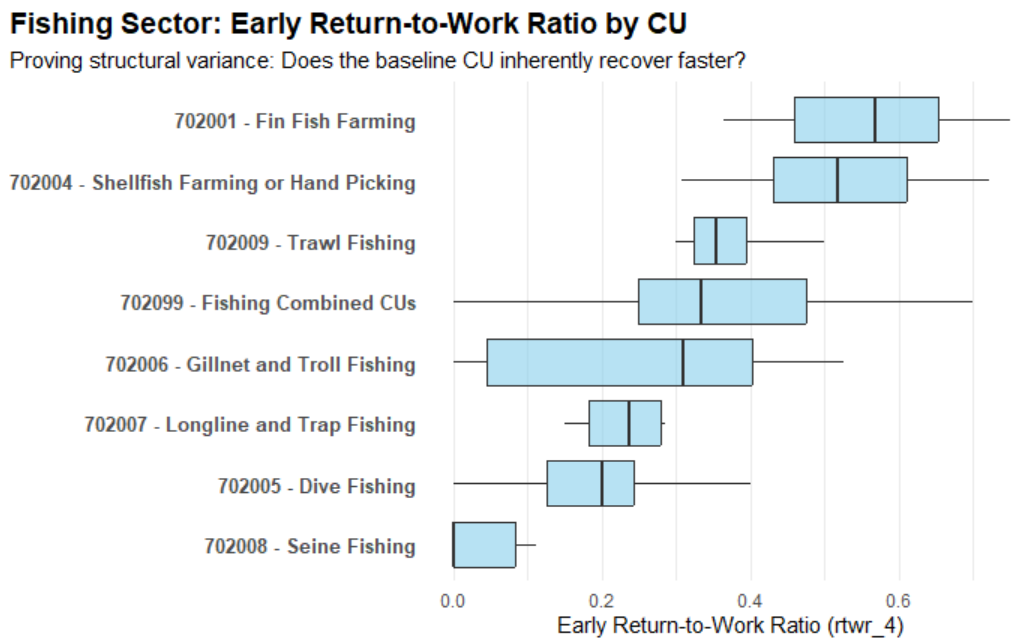

Figure C-7: Distribution of Early Return-to-Work Ratios Across Fishing Classification Units

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